

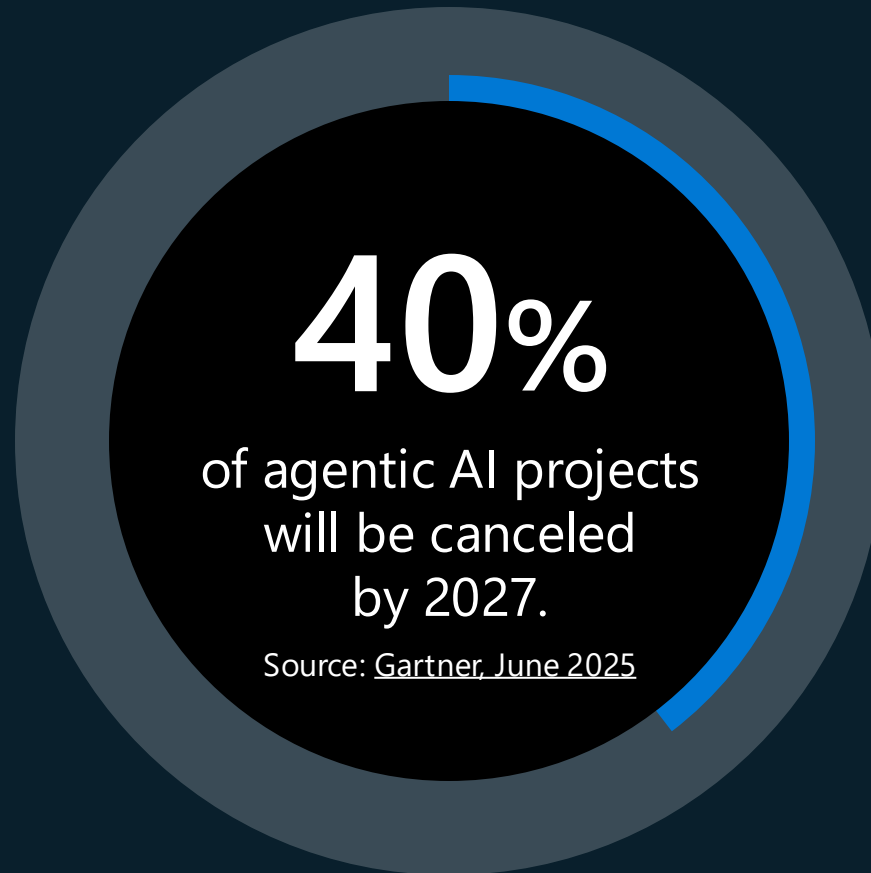


Introducing ACA Sandboxes

A New Primitive for Isolated,
Ephemeral, Production-grade
Compute on Azure

The future of work





Not because the models can't reason. Because the runtime can't keep up.

Five Places The Runtime Breaks



Budgets
that burn
unattended



Untrusted
code on a
developer's
laptop



Cold starts
that throttle
the loop



Workspaces
that die on
every restart



Tooling
stitched
together by
hand

Now Generally Available

Azure Container Apps Sandbox

<https://sandboxes.azure.com/>



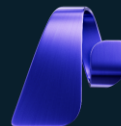
Fast, isolated and stateful compute infrastructure on demand

Execute Securely by Default
Sandbox isolation for any
untrusted workload

Resume Instantly
Preserve every interaction with
trusted enterprise controls

Burst to hyperscale
Spin up in sub-second time,
scale from zero to thousands,
pay nothing when idle

Foundation layer in



“

Atlas AI agents need to safely run code and work with files to answer complex questions in Cognite Data Fusion. Azure Container Apps Sandboxes provided secure, per-user isolation out of the box, letting Cognite focus on customer value instead of building infrastructure. The result is faster, trusted insights that help industrial customers go from days of investigation to cited answers in minutes.

Christian Flaschoff
Tech Lead, Atlas AI, Cognite



“

At Templafy we built an AI agent that helps our teams by doing longer-running source-code exploration on their behalf. Someone asks a question in a Slack thread, and behind the scenes the agent needs a real, isolated workspace where it can clone repositories, run tooling, and dig through code without touching anything it should not. It started as an engineer-facing tool for deep technical questions, but we recently opened it up to our product team for questions about undocumented product behavior.

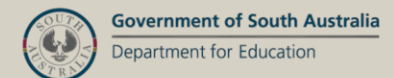
Nikoloz Buligini
Lead Frontend Engineer at Templafy



“

EdChat is an education AI platform where students learn safely with AI through coding, data analysis, and exploration. Azure Container Apps sandboxes would add built-in multi-tenant isolation and scale-to-zero, so each student could run in a cost-efficient, isolated environment with strong filesystem and network guardrails. Persistence and snapshots would let students resume work with files, code, and execution context intact -supporting longer-running tasks like notebooks and data exploration.

Cody Little
AI Technical Lead, Department for Education, South Australia



Use Cases for Sandboxes

Agent Workflows

Give AI agents persistent, isolated workspaces that survive across task boundaries

AI Code Execution

Safely run LLM-generated code in isolated environments with instant startup

Platform Building

Build your own platforms on top of the same primitive powering Microsoft services

Burst Workloads

Scale from zero to thousands of sandboxes in response to demand

Secure Multi-Tenant Compute

Strong isolation guarantees for running untrusted workloads from multiple tenants

Interactive User Sessions

Provide each user with their own isolated compute environment

Inside of the sandbox

Each sandbox runs as a hardware-isolated microVM with its **own Linux kernel, own virtual hardware, memory separation enforced by the CPU** virtualization.

- Your container runs exactly the way it always has. What changed is what's underneath it
- Sandboxes provide a different and **stronger boundary than a typical container runtime**, that shares the host kernel with everything else on the machine

Your code

your program, or whatever running harness agent just built

PER SANDBOX

Your container image

PER SANDBOX

Each sandbox has its own Linux kernel

PER SANDBOX

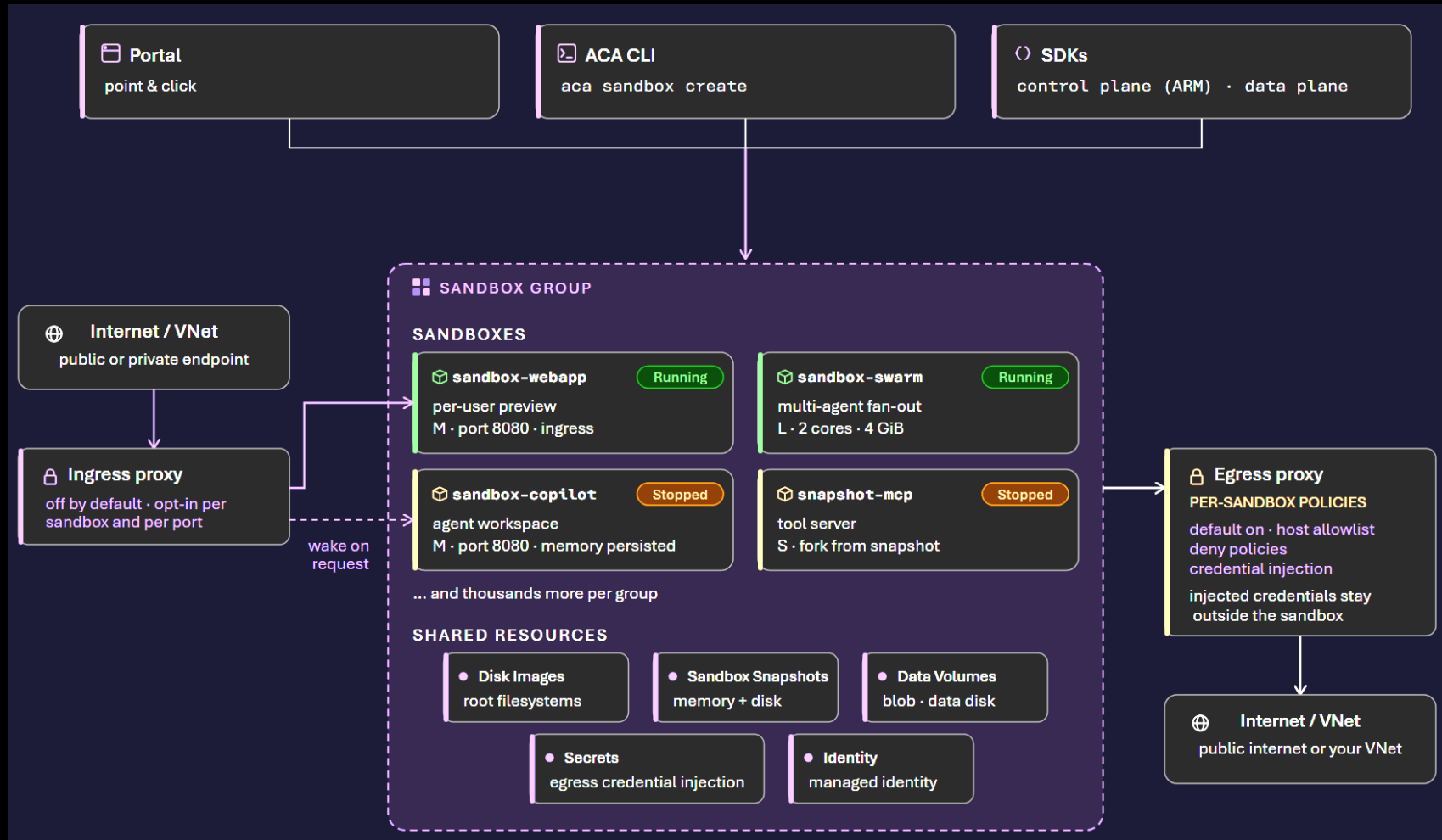


HARDWARE-ISOLATED MICROVM BOUNDARY

Enforced by the CPU (Intel VT-x / AMD-V), not by kernel permissions and not by policy.

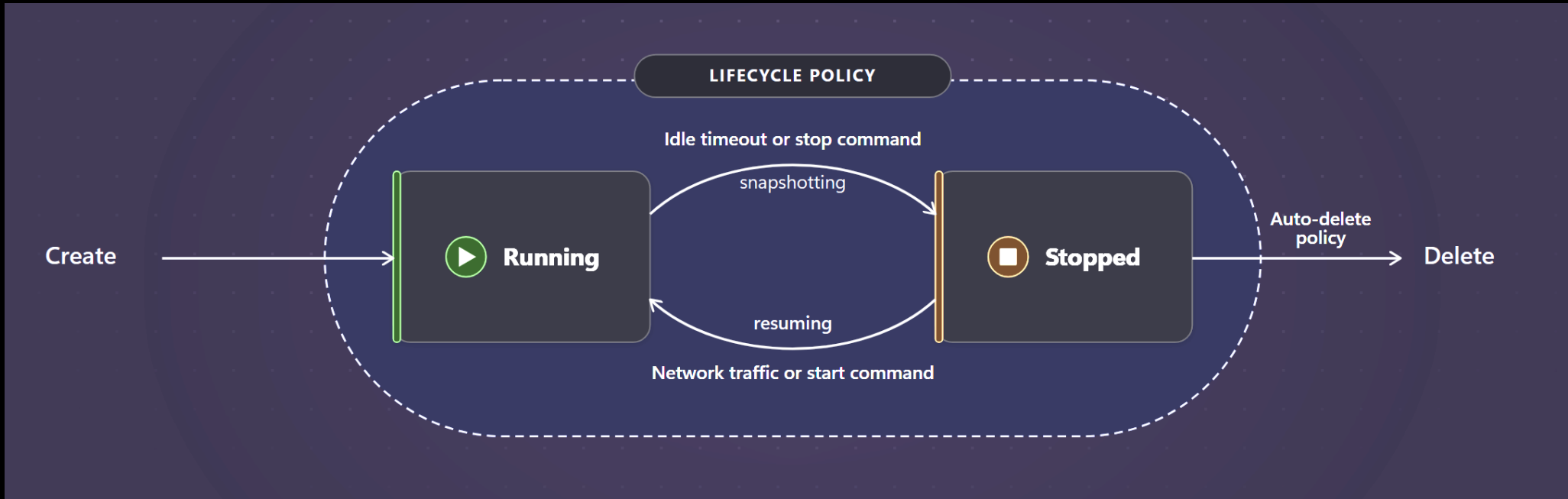
This is a boundary, not a guarantee about your code - anything you deliberately place inside a sandbox - credentials, mounted data, allowed network destinations - is available to whatever runs there.

Anatomy of a sandbox



- Sandbox group hosts thousands of sandboxes.
- Each sandbox boots from a disk image and can mount data volumes.
- Sandboxes are either Running or Stopped; stopped sandboxes incur no compute charges.
- Snapshots capture memory and disk for cloning or state restore.
- External access flows through an egress proxy with host allowlists, deny policies, and credential injection.
- Sandbox groups natively integrate with Azure virtual networks (VNETs) and Managed Identity (MI).

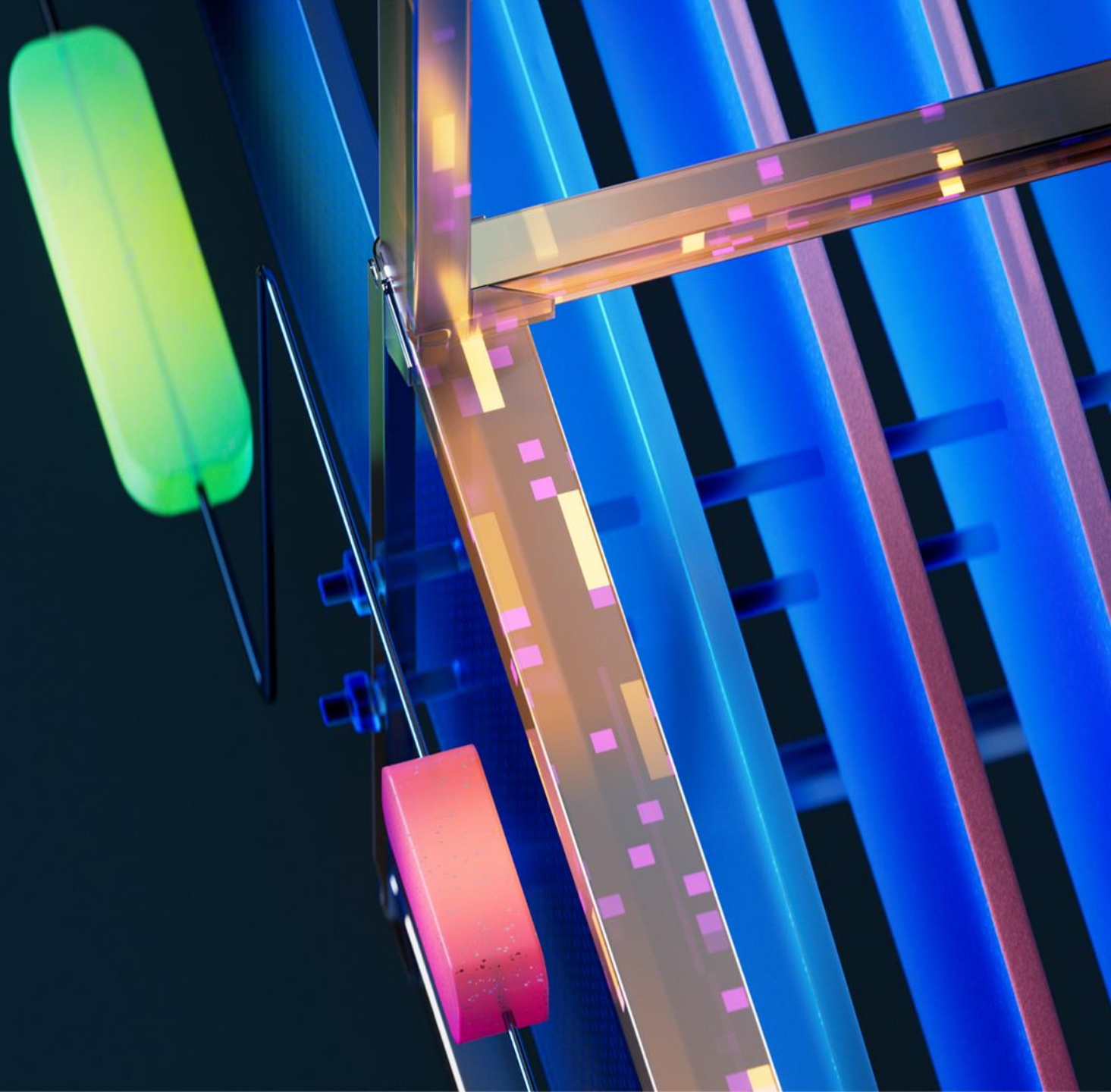
Sandbox Lifecycle



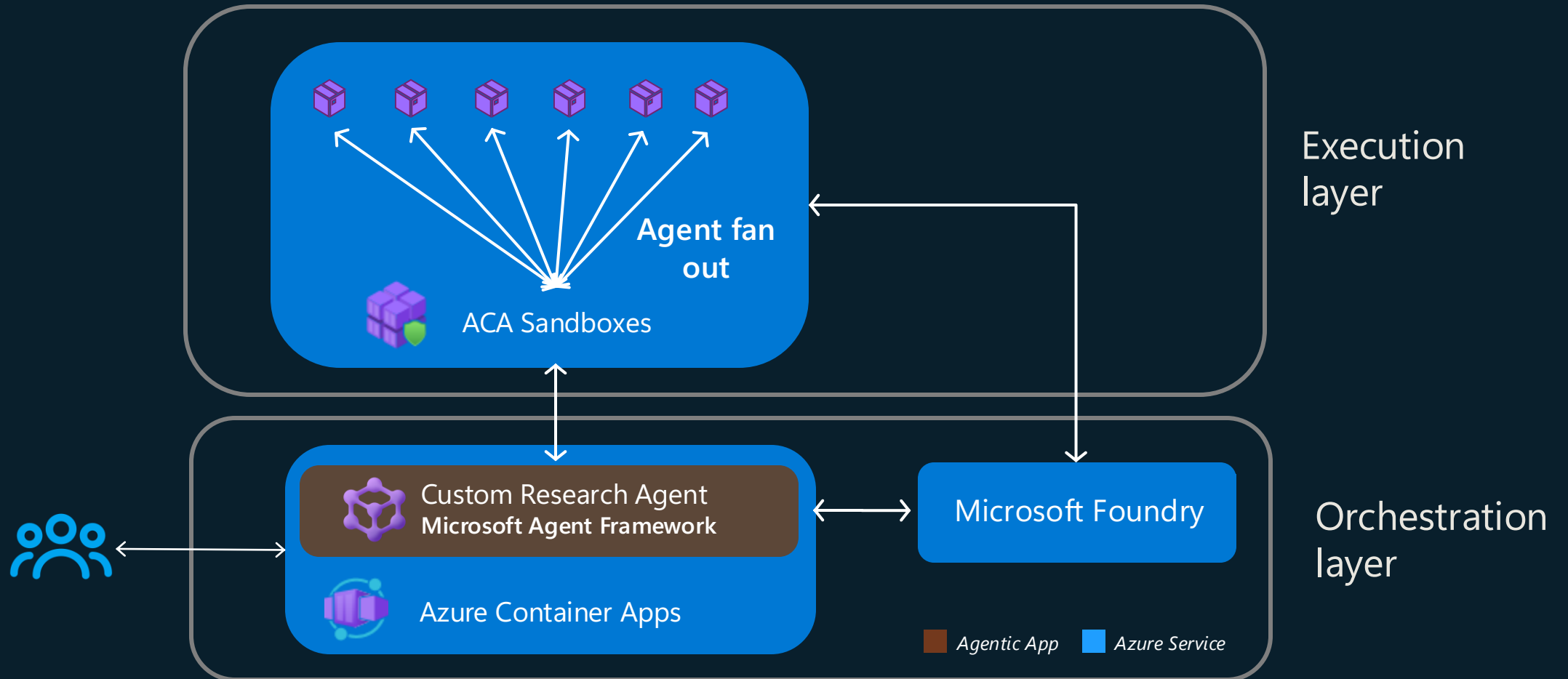
A sandbox has two core states: **Running** and **Stopped**. When sandbox is created it starts in running state. When sandbox is stopping, it snapshots disk and memory (optional) before it enters stopped state. While resuming it returns to running state under a second. A lifecycle policy can stop an idle sandbox and/or delete stopped sandbox after a set interval.

Sandboxes can enter a third state, **Disabled**. A disabled sandbox cannot start until it is re-enabled.

DEMOS



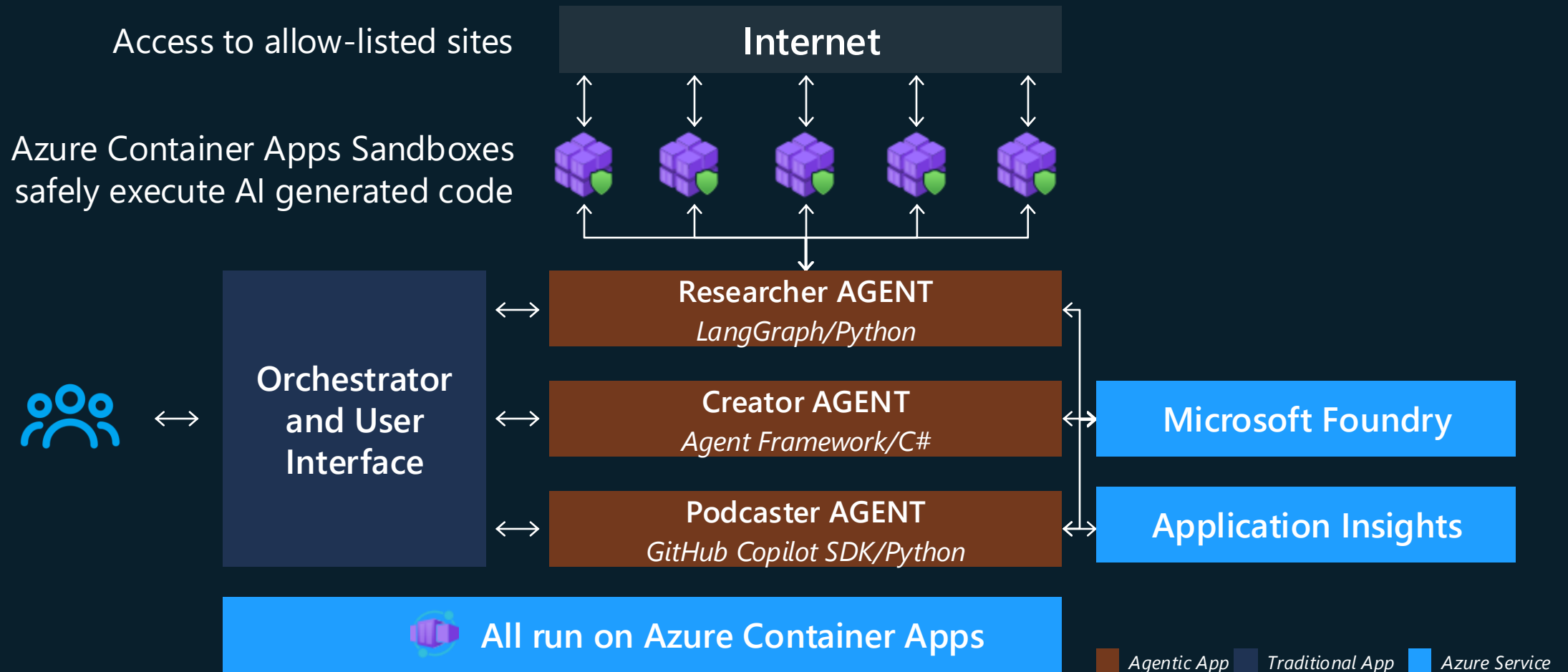
AI Research Agent Fan-out with ACA Sandboxes



Content Creator High-level Architecture



aka.ms/aca/build2026-dem312



USER-EXPERIENCE



Easy to use portal with embedded documentation

The screenshot displays the Azure Container Apps Sandboxes portal. At the top, there's a navigation bar with "Sign In — ACA Sandboxes" and a "Sign Up" button. The main header features the "Sandboxes" logo and the tagline "Isolated, on-demand microVMs — built for AI agents, code execution, and dev environments." Below this, there are "Get Started" and "View Docs" buttons. A central terminal window shows the following CLI commands and output:

```
CLI Python Agent Skill

INSTALL
> curl -fsSL https://aka.ms/aca-cli-install | sh

USE
> az login
✓ Signed in as user@contoso.com

> aca sandboxgroup create --subscription <sub-id> --resource-group <rg> --name my-group --location westus2 --set-config
✓ Sandbox group created

> aca sandbox create --disk ubuntu -l name=ubuntu-sandbox
✓ Sandbox created

ID                                STATE    CPU    MEMORY
c6cf7586-16e2-44ca-b671-d24fd569c0c3 Running  1000m  2048Mi
> aca sandbox exec -l name=ubuntu-sandbox -c "uname -a"
Linux adc-sandbox 6.12.8+ #1 SMP x86_64 GNU/Linux

> aca sandbox exec -l name=ubuntu-sandbox -c "echo \"build $(date -u +%Y%m%dT%H%M%S)\" > /sandbox/build.log"
✓ Saved /sandbox/build.log inside sandbox c6cf7586 (38 bytes)

> aca sandbox snapshot -l name=ubuntu-sandbox --name checkpoint-1
✓ Snapshot "checkpoint-1" created
```

At the bottom, a footer section highlights key features: "Sub-second startup. Boot, exec, throwaway in", "Strong isolation. Each sandbox runs in its own", and "Suspend and resume. Snapshot full memory +".

Create sandboxes in few clicks

Create Sandbox Group — ACA Sandbox

https://sandboxes.azure.com/sandbox-groups/create

ACA Sandboxes

Sandbox Groups / Create

Create Sandbox Group PUBLIC PREVIEW

Provision a new sandbox group to host your sandboxes

Simple and quick
Deploy in seconds with the essentials.

Advanced
Custom networking and MCP connector configuration.

BASICS

Name *
sandbox-group-af6a

Subscription *
Azure Subscription

Resource Group *
rg-sandboxes
⚠ A new resource group with that name will be created.

Region *
West US 2

FIRST SANDBOX (OPTIONAL)

☒ Create a sandbox now

Public disk image
ubuntu

Resource tier
M — 1 core / 2 GB / 20 GB (default)

Cancel Create

Settings
Azure Portal

Create sandbox under a second and start coding

Create

Sandboxes

GENERAL

Overview

RESOURCES

Sandboxes21

Disk Images1

Snapshots3

Volumes1

Secrets0

Connectors10

SETTINGS

Identity

Networking

Credentials

Settings

Azure Portal

Sandbox Groups / sandbox-group / Sandboxes

971343ac-123f-4122-b63b-10ae48fd3...RunningCopilotStop

Expose Port

bash

SessionsIssuesPull requestsGists

Copilot v1.0.80 uses AI.

Check for mistakes.

Tip: /sandbox

Open the sandbox settings, or enable/disable command sandboxing

https://docs.github.com/en/copilot/concepts/about-cloud-and-local-sandboxes

/Session: 0 AIC used

Loading: 1 instruction, 5 skills, 2 MCP serversGPT-5.6 Sol

Network Audit

50 allowed0 deniedUpdated 14:45:43Host

14:45:37POSTapi.enterprise.git...ALLOWED

14:45:37POSTapi.enterprise.git...ALLOWED

14:45:37GETlogin.microsoftonL...ALLOWED

14:45:37POSTapi.enterprise.git...ALLOWED

14:45:36POSTapp-17.westus2.log...ALLOWED

14:45:36GETapi.enterprise.git...ALLOWED

14:45:32POSTlogin.microsoftonL...ALLOWED

14:45:13POSTlogin.microsoftonL...ALLOWED

14:45:11POSTtelemetry.enterpri...ALLOWED

14:45:01POSTlogin.microsoftonL...ALLOWED

14:44:55POSTlogin.microsoftonL...ALLOWED

14:44:42GETlogin.microsoftonL...ALLOWED

CPU

1%Load: 0.79 / 0.22

Memory

7%537.7 MB / 8.0 GB

Network

Download25.7 B/sUpload86.5 B/sTotal Download546.3 MBTotal Upload2.5 MB

Storage

/containers/container-0/bundle/rootfs2.3 GB / 9.7 GB



ACA Sandboxes

Now Generally Available



Azure Container Apps Sandboxes

Now Generally Available